Akshat Sureshbhai Desai

Fullerton, CA | (714) 519-7380 | akshat.desai.754@gmail.com | LinkedIn | GitHub | Portfolio

EDUCATION

California State University, Fullerton | Fullerton, CA **Aug 2024 - May 2026**
Master of Science in Computer Science | *GPA: 3.65/4.0*
Charotar University of Science and Technology | Ahmedabad, India **Nov 2020 - May 2024**
Bachelor of Technology in Computer Science | *GPA: 3.6/4.0*

SKILLS

Languages: Python, C++, SQL, Bash
ML/AI: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, ONNX, ResNet, CNNs, Transfer Learning, Computer Vision, NLP, Grad-CAM, Time-Series Forecasting, Feature Engineering, Model Evaluation
LLM/GenAI: vLLM, LoRA, RAG, LangChain, LangGraph, FAISS, ChromaDB, Embeddings, Reranking, LLM Evaluation
MLOps/Cloud: Docker, Kubernetes, GCP Cloud Run, AWS, Deployment, Quantization, Inference Optimization
Systems/Data: Linux, HPC/SLURM, Git/GitHub, Flask, FastAPI, REST APIs, Pandas, NumPy, PyRadiomics, FreeSurfer

PROFESSIONAL EXPERIENCE

AI/LLM Research Developer | CSUF | Fullerton, CA **June 2025 - May 2026**

- Developed a production ***Verilog AI tutor*** for faculty and students to automate hardware design generation and simulation.
- Fine-tuned a ***32B code LLM*** on ***1,000 compiler-verified synthetic examples***, eliminating timing errors in Verilog testbenches.
- Reduced setup friction by building a browser-based simulator sandbox with Firebase login, chat history, compile logs, VCD downloads, and waveform previews without local Icarus Verilog/GTKWave setup.
- Reduced model footprint by ***70% by quantizing*** the fine-tuned model from ***64GB to 19.22GB***, enabling efficient vLLM inference.
- Improved Verilog generation quality using ***RAG and ChromaDB vector search*** over ***14,740 verified examples*** and theory docs.

Lead AI Developer | Worthwhile LLC | Escondido, CA **Oct 2025 - Jan 2026**

- Built ***Lightwall's local AI system*** for an interactive museum art installation, enabling real-time visitor conversations through Llama inference, speech recognition, radar sensing, and hardware responses.
- Automated installation behavior by converting LLM responses into structured ***hardware-control tags***, mapping visitor proximity and AI mood into LED commands, motor behavior, and personality-state changes.
- ***Reduced hallucinations*** in visitor-facing responses by adding project memory, live codebase scanning, grounded context injection, and JSON persona loading.

Machine Learning Research Associate | CSUF | Fullerton, CA **Nov 2024 - May 2025**

- Led Alzheimer's MRI radiomics research to classify ***CN, MCI, and AD*** patients from ***3D brain scans*** using machine learning.
- Increased usable cohort size from ***275 to 1,229 ADNI*** subjects by developing a custom ***"Subject Squashing" pipeline*** that reconstructed fragmented hippocampal and amygdala MRI subfield outputs into complete patient-level feature vectors.
- Reduced MRI ***processing runtime by 50%*** by automating FreeSurfer segmentation and PyRadiomics feature extraction across Linux HPC clusters using SLURM array jobs, Python scripting, and distributed **batch processing.**
- Achieved ***92.73%*** accuracy by reducing ***5,980 radiomics features to 77 biomarkers*** and training ***XGBoost*** with clinical scores.

Computer Vision & Imaging Intern | Space Applications Centre, ISRO | Ahmedabad, India **Dec 2023 - May 2024**

- Built a ***real-time satellite-camera exposure-control pipeline*** using Python, ***OpenCV***, pyueye, and IDS live camera streaming.
- Detected satellite-like objects in ***live camera streams using Haar Cascade/Object Tracking*****,** ROI extraction, and histogram analysis, then adjusted exposure with ***skewness-based control*** to correct under/overexposed frames in real time.
- Validated the algorithm in a ***dark-room setup*** using reflective satellite-like material against a space-like black background.

PROJECT EXPERIENCE

Solar Power Forecasting System | Python, LightGBM, Pandas, NumPy, Time-Series Forecasting

- Built a ***15-minute solar forecasting system*** using historical generation, NASA weather data, and LightGBM for energy planning.
- Cleaned ***210,000+ time-series*** records by resampling NASA weather to 15-minute intervals, interpolating sensor gaps, aligning power/weather timestamps, and capping outliers with ***Interquartile Range (IQR)***.
- Tuned LightGBM with **39** lag, time, and weather features, explaining ***96.02%*** of 2024 solar-output variation on ***unseen test data***.

Multi-Class Breast Cancer Subtype Classification | Python, PyTorch, ResNet, Grad-CAM, Medical Imaging

- Built an AI diagnostic pipeline to classify 8 breast cancer subtypes from ***7,909 biopsy*** images using ***PyTorch*** transfer learning.
- Trained an ImageNet-pretrained ***ResNet-50*** with preprocessing, augmentation, Adam optimization, and stratified splits, achieving ***91.54% accuracy***, ***99.68% AUC-ROC***, *and* ***98.67% specificity***.
- Outperformed prior reported methods by about ***5% accuracy*** and ***4% AUC-ROC***, while using ***Grad-CAM*** to highlight tumor-relevant regions for clinical interpretability.

CPP Dining AI Staffing Dashboard | AWS Hackathon | Flask, XGBoost, AWS Bedrock, LangChain, LangGraph

- Won the ***Cal Poly Pomona AWS Hackathon*** by building an AI dining-operations dashboard for campus dining services.
- Forecasted transactions and staffing demand across ***six dining roles*** using XGBoost, weather, dates, and campus events.
- Reduced projected annual labor waste by ***$52,000*** by optimizing daily worker allocation with machine learning forecasts.
- Integrated ***Flask APIs*** and an ***AWS Bedrock/LangChain/LangGraph*** chatbot for predictions and staffing insights.